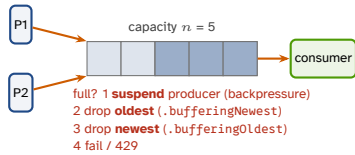
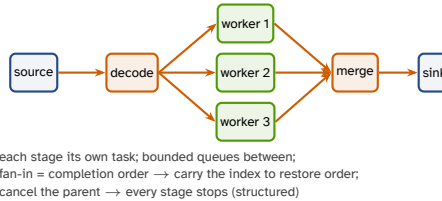


In one line: Every concurrent design answers two questions: **who may touch this state** (one owner + messages, or many + a lock) and **what happens when one side is faster** (a bounded buffer, backpressure, or dropping). The named patterns — producer-consumer, pool, actor, pipeline, fan-out/fan-in, reader-writer — are stock answers; Swift gives most of them a compiler-checked form (actor, TaskGroup, AsyncStream, Sendable).

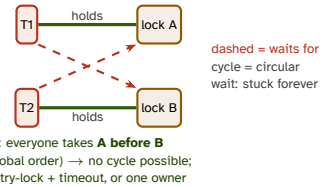
Producer-consumer, bounded buffer



Pipeline → fan-out → fan-in (TaskGroup)



Deadlock → lock ordering



Producer-consumer + thread pool

- A **bounded** buffer decouples rates. Unbounded = the slow side's problem becomes a memory problem. AsyncStream's default policy is **unbounded**; its `yield` never suspends, so it can only *drop*. Real backpressure (producer waits) = `AsyncChannel` (swift-async-algorithms), whose `send` suspends until consumed.
- AsyncStream has **one** consumer — concurrent iteration is a programmer error (SE-0314), never a broadcast. Fan-out to many = Combine subject or one stream per listener.
- Thread pool / executor**: a fixed set of workers pulls jobs from a queue — no per-job thread cost, no oversubscription. GCD: blocked workers make it spawn more (**thread explosion**). Swift's cooperative pool has $\approx$ one thread per core and **must never be blocked** (`DispatchSemaphore.wait`, sync I/O) — it cannot grow, so it starves.

Example — fan-out/fan-in, buffered stream

```
func thumbs(_ urls: [URL]) async throws -> [UIImage] {
    try await withThrowingTaskGroup(of: (Int, UIImage).self) { g in
        var out = [UIImage?](repeating: nil, count: urls.count)
        for (i, url) in urls.enumerated() {
            if i > 3, let r = try await g.next() { out[i.0] = r.1 } // max 4 in flight
            g.addTask { (i, try await load(url)) } // fan-out
        }
        for try await r in g { out[r.0] = r.1 } // fan-in; a throw cancels the rest
        return out.map { $0! } // input order restored
    }
}

let (events, cont) = AsyncStream.makeStream(of: Event.self,
    bufferingPolicy: .bufferingNewest(100)) // drops OLDEST
mqtt.onMessage = { cont.yield($0) } // producer never blocks
Task { for await e in events { await store.apply(e) } } // one consumer
```

Futures/promises vs async/await

A **future** is a read-only handle to a value that arrives later; the **promise** is its write side. Chained with `then`/combinators — control flow and errors move into closures. **async/await** is the same thing written as straight-line code: `try/catch`, loops, `defer` work again; a `Task` is a future (`await task.value`). Combine's `Future` runs its closure **at creation** (eager) and emits once — wrap in `Deferred` to make it lazy.

Actor model

Hewitt 1973, Erlang, Akka: an actor owns its state, has a **mailbox**, handles **one message at a time**, and talks only by messages → no locks, no shared state. Swift actor: same isolation, compiler-enforced, but **not FIFO** (jobs run by priority) and **reentrant**: at each `await` inside it the next message runs, so re-check state after suspension (timeline on *concurrency*). An Erlang process is not reentrant — a synchronous call to itself would hang forever, so OTP's `gen_server:call` exits with `calling_self` instead.

Read-write lock

Many readers or one writer (pthread_rwlock; GCD concurrent queue + `.barrier` writes). Pays off only for long, contended reads; else it costs more than a mutex, and writers can **starve**. Replaced by a serial queue / actor (simple), `os_unfair_lock/Mutex` (fast), or **copy-on-write** snapshots (readers take a value, the writer swaps it).

Double-checked locking

if x == nil { lock; if x == nil { x = make() } } — the unlocked read races; without acquire/release ordering a reader can see the pointer before the object's fields. Swift: `static let / global let` run **exactly once**, thread-safe (`swift_once`); lazy `var` is **not** thread-safe.

Shared memory + locks vs messages

	shared memory + locks	immutability + messages
correctness	every access must remember the lock	ownership makes races impossible
failure mode	data race, deadlock, inversion	mailbox growth, stale copies
cost	uncontended lock is cheap	copies (CoW), actor hops
Swift	Mutex (iOS 18), <code>os_unfair_lock</code> , <code>NSLock</code>	actor, Sendable values, AsyncStream

Default to ownership + messages; keep a lock for a tiny hot critical section with no await inside (never hold a lock across a suspension).

Deadlock rules

Needs all four Coffman conditions: mutual exclusion, hold-and-wait, no preemption, **circular wait**. Break one: global lock order; try-lock with timeout; take everything at once; one owner. Never call out (`delegate`, `callback`, `main.sync`) while holding a lock.

Lock-free / CAS

Read, compute, `compareExchange(expected:desired:)`; retry if someone got there first — no blocking, but livelock under contention and the **ABA** problem. Swift 6: `Atomic<Int>` in `Synchronization` (iOS 18), e.g. `wrappingAdd(1, ordering: .relaxed)`. Use for counters and flags; not for data structures.

Reactive streams + backpressure

Subscriber *pulls*: it signals **demand** (`Subscribers.Demand.max(n)`) and the publisher may send no more than that. `sink/assign request.unlimited` — no backpressure. Bound it with `flatMap(maxPublishers: .max(3))`, `buffer(size:prefetch:whenFull:)`, or a custom subscriber.

Structured concurrency

Child tasks (`async let`, `TaskGroup`) **cannot outlive** their scope: the parent waits, **cancellation flows down**, **errors flow up**, task-locals are inherited. Cancellation is cooperative (`checkCancellation`, `withTaskCancellationHandler` for callbacks). `SwiftUI.task` ties a task to a view's lifetime. `Task {} / Task.detached` are escape hatches that leak.

Traps

- ! `DispatchSemaphore.wait()` in async code — blocks a pool thread; can deadlock the pool.
- ! "Actors make it atomic" — only between awaits.
- ! AsyncStream is not a broadcast, and unbounded by default.
- ! A task group returns in *completion* order, not input order.

Remember

One owner or one lock. Bound every buffer. Children die with the parent.